

E-Commerce Business Analytics Report

Key Findings, Trends, Anomalies & Data-Driven Recommendations

Qwetrum Technologies - Data Analysis Internship - Capstone Project

1. Executive Summary

This report analyses the performance of an e-commerce business using a dataset of 10,000 orders placed by 2,000 customers across 1,200 products and four data tables. The analysis combines SQL-based metric extraction, statistical hypothesis testing, and interactive dashboarding to turn raw transaction data into clear, actionable business insight.

The headline picture is healthy but with clear opportunities. Completed orders generated \$6.94 million in revenue and \$2.63 million in profit, at an average order value of \$914.51. However, roughly one in four orders did not complete — cancelled or returned — representing a substantial recoverable opportunity. Returns are concentrated in specific product categories, a pattern confirmed as statistically significant.

Key Metrics at a Glance

Metric	Value
Completed Revenue	\$6,943,887
Total Profit	\$2,627,747
Completed Orders	7,593
Average Order Value	\$914.51
Overall Profit Margin	~38%
Non-Completed Orders	24.1% of total

2. Methodology & Data Overview

This analysis draws on a relational e-commerce dataset comprising four linked tables, which together capture the full customer-to-transaction journey. Working across multiple related tables — rather than a single flat file — allows revenue, profitability, and customer behaviour to be analysed at the appropriate level of detail.

2.1 Data Structure

Table	Rows	Description
customers	2,000	Customer profile, location, and segment
products	1,200	Product catalogue with cost and list price
orders	10,000	Order header: date, channel, status
order_items	~17,500	Line items: product, quantity, unit price

The tables are linked by key relationships: each order belongs to a customer, and each order is composed of one or more line items, each referencing a product. Revenue is calculated at the line-item level as quantity multiplied by unit price, and profit subtracts each product's cost price.

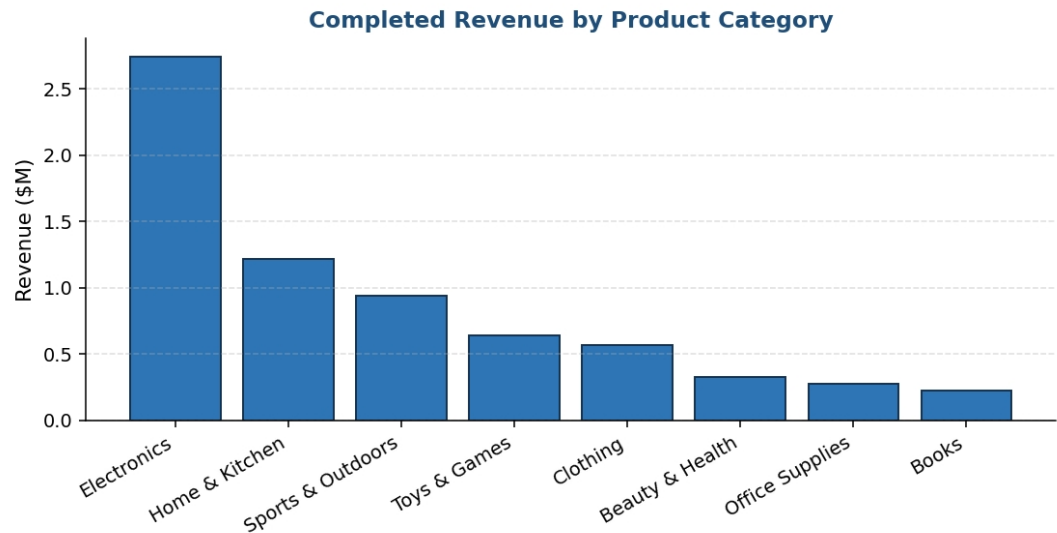
2.2 Analytical Approach

- SQL was used to extract and aggregate key business metrics across the joined tables.
- A Chi-Square statistical test was applied to determine whether observed differences are significant or simply random variation.
- An interactive Power BI dashboard was built to allow management to explore the findings dynamically.
- Throughout, the focus is on completed orders for revenue figures, with cancelled and returned orders analysed separately as a loss category.

3. Key Findings

3.1 Revenue is concentrated in a few categories

Electronics is the dominant revenue driver, generating \$2.74 million — more than double the next category, Home & Kitchen (\$1.22 million). The top three categories (Electronics, Home & Kitchen, Sports & Outdoors) account for the majority of completed revenue.



3.2 Profit margins are stable across categories

Profit margins are remarkably consistent, sitting between 37% and 39% in every category. This means revenue volume — not margin efficiency — is what separates strong categories from weak ones. There is no single “high-margin” category to prioritise; growth comes from selling more, not from a pricing advantage in any one area.

3.3 Sales channels are well balanced

Web is the leading channel (\$3.05M), followed closely by the Mobile App (\$2.85M), with Marketplace a smaller contributor (\$1.04M). Notably, average order value is very similar across all three channels (\$898–\$929), so the difference between channels is one of customer volume rather than basket size.

Channel	Orders	Revenue	Avg Order Value
Web	3,398	\$3,052,166	\$898.22
Mobile App	3,074	\$2,850,151	\$927.18
Marketplace	1,121	\$1,041,570	\$929.14

4. Trends

4.1 Revenue is steady rather than seasonal

Month-over-month revenue across the 2023–2025 period is stable, fluctuating within a consistent band without strong seasonal peaks or a pronounced growth trajectory. For the business, this means demand is dependable and predictable — useful for planning — but also that there is no organic seasonal surge to rely on; growth must be actively driven.

4.2 The Consumer segment dominates

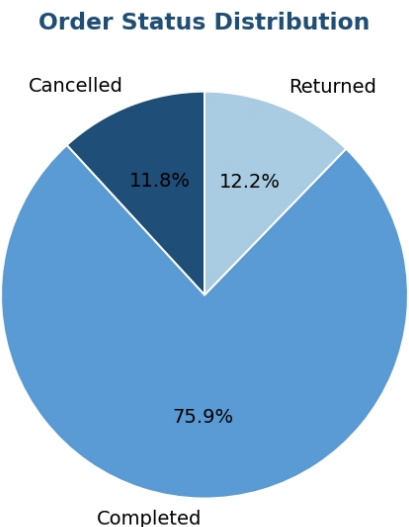
Across every country, the Consumer segment is the largest revenue contributor, led by Consumer-USA (\$630K), Consumer-Japan (\$478K), and Consumer-UK (\$450K). Corporate and Home Office segments trail well behind, indicating the business is fundamentally consumer-driven.

Segment	Country	Revenue
Consumer	USA	\$630,296
Consumer	Japan	\$478,452
Consumer	UK	\$450,393
Consumer	India	\$440,130
Consumer	Germany	\$382,351

5. Anomalies

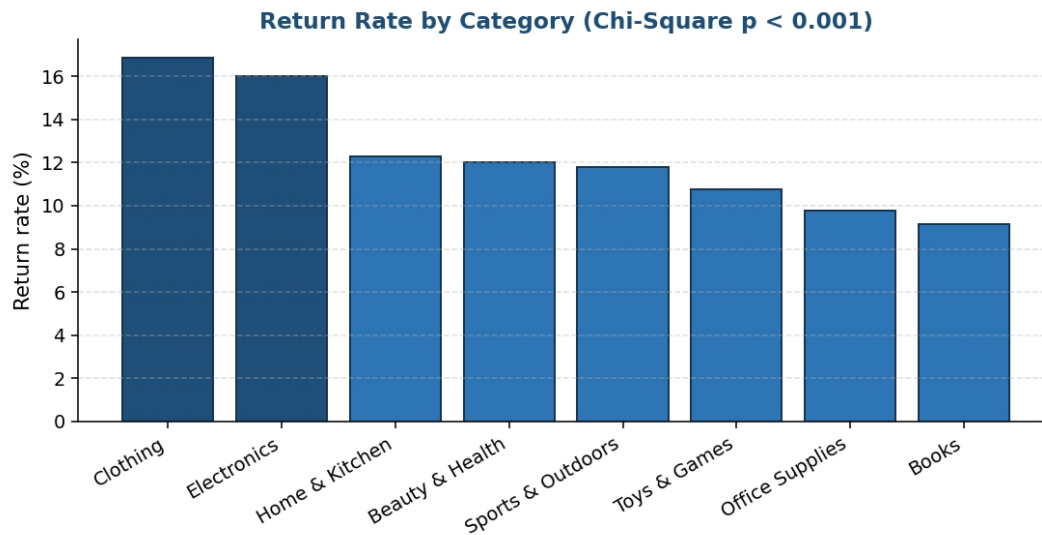
5.1 A quarter of all orders never convert

The single most important anomaly: 24.1% of orders are either cancelled (11.8%) or returned (12.2%). Together these non-completed orders represent roughly \$2.45 million in revenue that did not convert cleanly. This is the largest single opportunity surfaced by the analysis.



5.2 Returns are not uniform — they cluster by category

A Chi-Square test of independence confirmed that return rates differ significantly across product categories ($\chi^2 = 111.19$, $p < 0.001$). Clothing (16.9%) and Electronics (16.0%) are returned nearly twice as often as Books (9.2%) and Office Supplies (9.8%). Because this difference is statistically significant, it reflects a genuine pattern, not random variation.



6. Data-Driven Recommendations

6.1 Tackle the return problem where it concentrates

- Prioritise Clothing and Electronics for returns reduction — these two categories drive the highest return rates and together represent the clearest recoverable revenue.
- For Clothing, invest in better sizing guides, fit information, and product imagery, since fit uncertainty is the typical driver of apparel returns.
- For Electronics, review product descriptions, quality control, and post-purchase support to reduce defect- and expectation-driven returns.

6.2 Protect and grow the core

- Double down on Electronics as the revenue engine, while recognising that all categories share a similar ~38% margin — so volume growth, not margin optimisation, is the lever.
- Nurture the Consumer segment, which dominates revenue in every market, through loyalty and retention programmes.

6.3 Convert more orders

- Investigate the cancellation pathway: an 11.8% cancellation rate suggests friction at checkout, payment, or delivery-expectation stages worth diagnosing.
- Because channels perform similarly on order value, invest in whichever channel is cheapest to grow by volume rather than assuming one is inherently more valuable.

7. Limitations

A complete analysis acknowledges its boundaries. The following limitations should be considered when acting on this report:

- The dataset is synthetic, generated for the capstone project. The figures demonstrate analytical methodology rather than reflecting a real market, and would be re-validated against live business data before driving real decisions.
- The analysis covers transactional data only. It excludes marketing spend, acquisition cost, seasonality, and return reasons — all of which would be needed for a full growth and profitability picture.

8. Conclusion

The business is profitable and stable, with Electronics and the Consumer segment as its foundation. The clearest path to improvement is operational rather than strategic: recovering even a fraction of the 24% of orders that cancel or return — especially the statistically-confirmed return hotspots in Clothing and Electronics — would translate directly into recovered revenue without acquiring a single new customer.